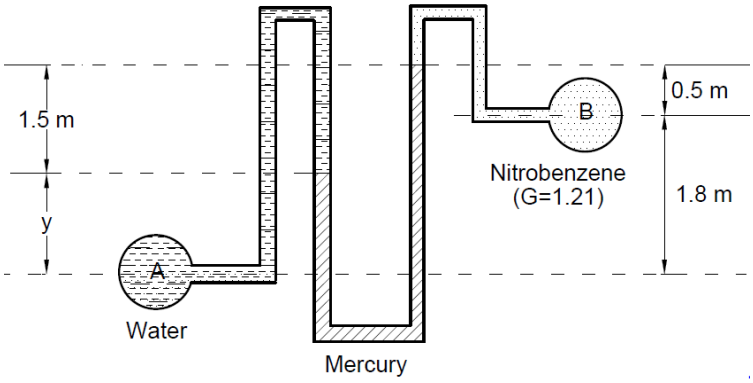
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**CORE**
Course Name	**Fluid Mechanics**	**Course Code**	**CIE 2121**
Semester	**III**	**Max Marks**	**50**
Date	**20/11/2024**	**Duration**	**3 hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A smooth steel sleeve of length 30cm and internal diameter 5.075cm slides down concentrically along a vertical solid steel shaft of 5cm diameter. The angular space between the sleeve and the shaft is filled with oil of viscosity 6.83 poise. The steady velocity of the sleeve is 2.5m/min. Calculate the force with which the sleeve moves downward?	**3**	1	1,2	4
1B	Discuss cohesion and adhesion in connection to surface tension and capillarity.	**2**	1	1,2	2
1C	Two pipes carrying water and nitrobenzene (specific gravity 1.21) are connected by a mercury differential gauge as shown in the figure. Evaluate the pressure difference in the two pipes in kPa. 	**5**	2	1,2,12	4
2A	Distinguish between (i) Steady and unsteady flow (ii) Uniform and non-uniform flow	**4**	3	1,2,12	2
2B	A circular tank of 3m diameter contains water up to a height of 4m. The tank is provided with an orifice of diameter 0.3m at the bottom. Determine the	**3**	3	1,2,12	3

	time taken by water (i) to fall from 4m to 1m (ii) for completely emptying the tank. Take $C_d = 0.6$.				
2C	A pipe of diameter 300mm carries water from point A to point B. The rate of flow is 1.5 cumecs. The pressures at the points A and B are given as 294.3 kN/m^2 and 225.3 kN/m^2 respectively, while the datum head at A and B are 25m and 30m. Calculate the loss of head between A and B.	**3**	3	1,2,12	4
3A	A laminar flow is taking place in a pipe of diameter of 0.4m. The maximum velocity is 2m/s. Determine the mean velocity and the radius at which this occurs. Also, compute the velocity at 0.006m from the wall of the pipe.	**4**	4	1,2,3,6,12	**3**
3B	Apply Bernoulli's equation at two sections of a uniform diameter pipe to determine the expression for loss of energy due to friction (Darcy-Weisbach Equation)	**4**	4	1,2,3,6,12	3
3C	Illustrate any four laws of friction for Laminar flow.	**2**	4	1,2,3,6,12	2
4A	Two reservoirs whose water surface elevations differ by 12m are connected by the following horizontal compound pipe system starting from the high level reservoir. The length, diameter and friction factor of the pipes are as follows; $L_1 = 200$m, $D_1 = 0.2$m, $f_1 = 0.032$ and $L_2 = 500$m, $D_2 = 0.3$m, $f_2 = 0.024$. Considering all losses and assuming that all changes of section are abrupt, calculate the discharge through the system.	**5**	4	1,2,3,6,12	4
4B	In a laboratory experiment, a Cippoletti weir having a crest width of 30 cm is used to measure the flow of water in a rectangular channel 50cm wide and 40cm deep. If the water level in the channel is 20cm above the weir crest, determine the discharge in the channel considering velocity of approach. Take $C_d = 0.6$	**3**	5	1,2,6,12	3
4C	Determine the critical depth, critical velocity and minimum specific energy for a flow of 12m^3/s in a rectangular channel of width 3.5m.	**2**	5	1,2,6,12	3
5A	Determine the conditions required for a rectangular channel section such that it offers the least resistance to flow and hence it can carry maximum discharge for a given slope, area and roughness.	**4**	5	1,2,6,12	3
5B	Explain the applications and places of occurrence of hydraulic jump.	**3**	5	1,2,6,12	2
5C	A trapezoidal channel has a bed width of 5m, side slope of 1V:1.5H and Manning's n =0.015. Calculate the channel bed slope as it conveys discharge of 60 cumecs for a depth of flow of 2m. Comment on the state of flow?	**3**	5	1,2,6,12	4

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO	Total Marks
1	5
2	5
3	10
4	15
5	15

BL	Total Marks
2	**11**
3	**20**
4	**19**
5	**---**